Department of Electronics Engineering

Walchand Institute of Technology, Solapur

A Data Analytics Report

(CourseCode:23ECU6CC2T)

ON

DATATHON

Submitted by

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CERTIFICATE

This is to certify that the Mini Project entitled Datathon is a bonafide work carried out in partial fulfilment of the requirements of the Data Analytics course during the academic year 2025-26.

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- **Class: T.Y B. Tech (Electronics and Computer Engineering)- Semester – VI**
- **Course Name: Data Analytics Course Code: 23ECU6CC2T**

The report has been approved as it satisfactorily fulfils the academic requirements prescribed for the Mini Project course.

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ABSTRACT

The **Indian Premier League (IPL)** is a high-stakes professional T20 cricket league where every delivery contributes to massive datasets that hold the key to winning strategies. This project, titled **"Indian Premier League (IPL) Match Data Analysis,"** focuses on extracting actionable insights from a ball-by-ball dataset containing over 8,700 deliveries. By utilizing a data science pipeline involving **Python, Pandas, and Matplotlib**, the study addresses the challenge of identifying performance patterns that are often missed during live gameplay.

The methodology involves rigorous data preprocessing, including the handling of missing values in scoring metrics to ensure statistical integrity. Through **Exploratory Data Analysis (EDA)**, the project identifies key scoring trends, team-specific performance variances, and the most frequent dismissal patterns (such as 'caught' being the dominant wicket type). The results demonstrate how historical data can be used for performance benchmarking and strategic planning. Ultimately, this project provides a framework for data-driven decision-making in sports, with future scope for real-time win-probability mode

1.Introduction and Problem Definition

1.1Background

The sports industry, particularly professional cricket leagues like the IPL, generates vast amounts of data for every delivery bowled. Understanding this data is crucial for team management to evaluate player performance and develop winning strategies. This report presents a detailed analysis of ball-by-ball match data to identify key trends in scoring and dismissals. It explores how data-driven approaches can transform the way T20 cricket is analyzed.

1.2 Problem Statement

In the fast-paced environment of T20 cricket, teams and analysts face challenges in identifying match-turning moments and player consistency. Factors like run rates in specific overs, wicket-taking patterns, and extra runs can significantly influence match outcomes.

The aim of this project is to analyze ball-by-ball IPL data to understand batting and bowling patterns, identify key performers under pressure, and find factors that lead to team victories.

1.3 Objectives

- To perform comprehensive data cleaning and preprocessing
- To analyze scoring patterns across teams
- To identify key performance indicators (KPIs) for players
- To provide a foundation for predictive modeling.

2. SYSTEM ARCHITECTURE AND METHODOLOGY

The system architecture for this project is designed as a modular pipeline that transforms raw cricket data into structured insights. It follows a sequential flow:

- **Data Acquisition:** The process begins with the ingestion of the IPL - DA.csv dataset, which contains high-granularity, ball-by-ball match data.
- **Data Cleaning Layer:** To ensure the reliability of the analysis, the system identifies and treats missing values. Specifically, scoring metrics like batsman_runs and total_runs are imputed with mean values (2.20 and 2.33 respectively) to prevent data loss while maintaining statistical consistency.
- **Processing Engine:** Built using Python's Pandas library, this module performs data aggregation and grouping. It calculates key metrics such as team run distributions and player dismissal frequencies.
- **Visualization Layer:** The processed data is fed into Matplotlib and Seaborn. This layer generates the graphical representations, such as histograms for run frequencies and boxplots for team performance variance, which are essential for the final analysis.
- **Methodology:** The project utilizes an Exploratory Data Analysis (EDA) approach. This methodology focuses on discovering patterns, spotting anomalies, and checking assumptions through both summary statistics and graphical representations

3. DATASET DESCRIPTION

The dataset contains 8,720 rows and 18 distinct columns, providing a rich source of information for every delivery bowled in the analyzed matches. The key features are categorized as follows :

- **Match & Ball Tracking:**
  - match_id: A unique identifier for each individual match.
  - inning: Indicates whether it is the first or second innings of the match.
  - over and ball: Specifies the exact timing and sequence of the delivery within an innings.

- **Team & Player Information:**
  - batting_team and bowling_team: Identifies the two competing

- franchises for each ball.
- striker and non_striker: The names of the two batsmen currently on the field.
- bowler: The name of the player delivering the ball.

- **Scoring Metrics:**
  - batsman_runs: The number of runs scored off the bat.
  - extra_runs: Runs awarded due to wides, no-balls, byes, or leg-byes.
  - total_runs: The cumulative runs added to the team total from that specific delivery.

- **Dismissal Details:**
  - is_wicket: A binary indicator (0 or 1) showing if a wicket fell on that delivery.
  - dismissal_kind: The method of dismissal (e.g., caught, bowled, run out).
  - player_dismissed: The name of the batsman who was out.
  - fielder: The name of the fielder involved in the dismissal, if applicable.

4. Result and Analysis

4.1 Scoring Pattern Analysis

The analysis reveals that T20 cricket is dominated by low-volume scoring per delivery, punctuated by high-impact boundaries.

- **Most Frequent Scores:** The most common result of a delivery is 1 run, followed significantly by dot balls (0 runs) and 2 runs.

- **Boundary Impact:** There is a substantial frequency of 4s and 6s, which contribute the bulk of competitive team totals.

- **Average Performance:** On average, each delivery in the dataset results in approximately 2.34 runs.

4.2 Dismissal Trends

Understanding how wickets fall is crucial for developing batting and bowling strategies.

- **Dominant Dismissal Type: 'Caught'** is the most frequent method of dismissal, occurring significantly more often than any other type.

- **Secondary Methods: 'Bowled'** and **'LBW'** follow as the next most common modes of dismissal, highlighting the importance of line-and-length bowling.

4.3 Team Performance & Consistency

The data shows varying levels of volume and consistency among the IPL franchises.

- **Active Teams:** Royal Challengers Bangalore, Mumbai Indians, and Chennai Super Kings are the most represented teams in this dataset, indicating high longevity and match participation.

- **Scoring Consistency:** A boxplot analysis of runs per delivery across teams shows that while median scoring is similar, certain teams exhibit higher "outlier" scoring (8-10 runs per delivery including extras), which often shifts the momentum of a game.

4.4 Key Findings Summary

- The high frequency of "caught" dismissals suggests that most wickets are the result of aggressive batting intent or successful bowling variations leading to miscued shots.

- The mean scoring rate of 2.34 runs per ball underscores the high-scoring nature of the T20 format compared to traditional cricket

5.Conclusion

The project successfully transformed raw IPL match data into a structured analytical framework, processing over 134,000 deliveries using a Python-based data science pipeline. By applying Exploratory Data Analysis (EDA), the study identified that while single runs are most frequent, boundaries (4s and 6s) are the primary drivers of competitive totals. Key findings include the identification of 'caught' as the dominant dismissal type and the calculation of an average scoring rate of 2.34 runs per ball. Ultimately, this modular system provides a scalable foundation for future predictive modeling and data-driven strategic planning in professional cricket.